Basic Electric Circuits

Today's Topics:

- Linearity And Superposition
- Thevenin's and Norton's Theorems

Important!

- The first lab will be at 5th Week!
- The second lab will be at 7th Week!

Other labs will be after midterm.

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Let's start with:

- Linearity And Superposition

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Linearity and Superposition: Linearity

Basically, an equation is said to be linear in Circuits, Signals and Systems (CSS).
if the following properties hold:

- **homogenity**
- **additivity**

What does this mean? We first look at the property of homogenity.

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Linearity : Homogeneity.

Homogeneity means that scaling the input of a system by a constant results in the output being scaled by the same constant.

The equations that we think are linear in CSS, turn out not be linear because they fail the homogenity property. We next consider such an example.

Illustration: Does homogenity hold for the following equation?

Given,

$$y = 4x \qquad \text{Eq 3.1}$$

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Linearity : **Homogeneity (scaling).**

Illustration: **Does homogenity hold for the following equation?**

Given,

$$y = 4x \qquad \text{Eq 3.1}$$

If x = 1, y = 4. If we double x to x = 2 and substitute this value into Eq 3.1 we get y = 8.

Now for homogenity to hold, scaling should hold for y. that is, y has a value of 4 when x = 1. If we increase x by a factor of 2 when we should be able to multiply y by the same factor and get the same answer and when we substitute into the right side of the equation for x = 2.

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Linearity : **Homogeneity (scaling).**

Illustration: Does homogenity hold for the following equation?

Given,

$$y = 4x + 2 \qquad \text{Eq 3.2}$$

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Linearity : Homogeneity (scaling).

Illustration: Does homogenity hold for the following equation?

Given,

$$y = 4x + 2 \qquad \text{Eq 3.2}$$

If x = 1, then y = 6. If we double x to x=2, then y = 10.

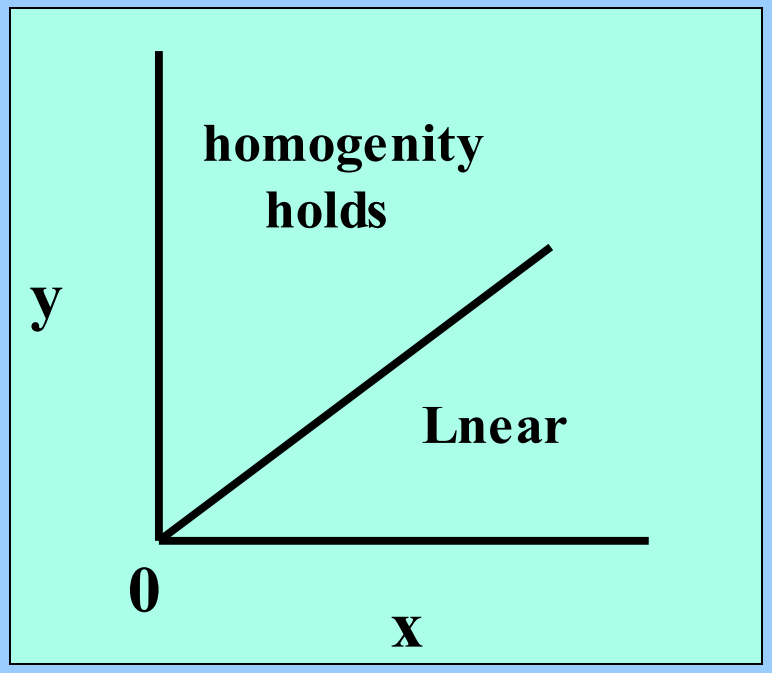
Now, since we doubled x we should be able to double the value that y had when x = 1 and get y = 10. In this case we get y = (2)(6) = 12, which obviously is not 10, so homogenity does not hold.

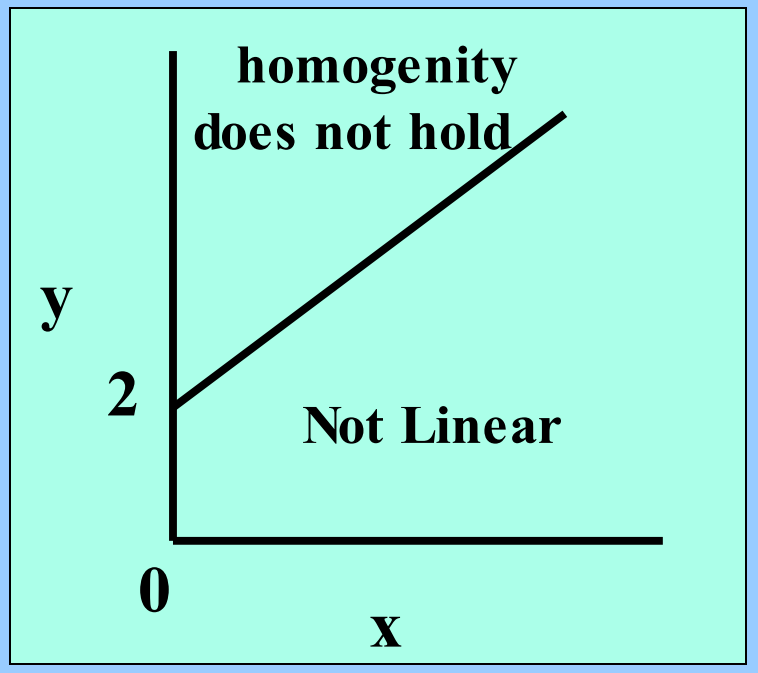
We conclude that Eq 3.2 is not a linear equation in CSS. In some ways that goes against the gain of what we have been taught about linear equations.

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Linearity : Homogeneity (scaling).

Many of us were brought-up to think that if plotting an equation yields a straight line, then the equation is linear in CSS. From the following illustrations we have;





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Linearity : **Additivity Property.**

Additivity means that the system's response to a sum of inputs equals the sum of its responses to each input applied separately.

This can be explained by considering the following illustrations.

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Linearity : Additivity Property.

Illustration: Given, $y = 4x$.

Let $x = x_1$, then $y_1 = 4x_1$
Let $x = x_2$, then $y_2 = 4x_2$

Then $y = y_1 + y_2 = 4x_1 + 4x_2$ Eq 3.3

Also, we note,

$$y = f(x_1 + x_2) = 4(x_1 + x_2) = 4x_1 + 4x_2 \quad \text{Eq 3.4}$$

Since Equations (3.3) and (3.4) are identical, the additivity property holds.

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Linearity : Additivity Property.

Illustration: Given, $y = 4x + 2$.

Let $x = x_1$, then $y_1 = 4x_1 + 2$
Let $x = x_2$, then $y_2 = 4x_2 + 2$

Then $y = y_1 + y_2 = 4x_1+2 + 4x_2+2 = 4(x_1+x_2) + 4$ Eq 3.5

Also, we note,

$$y = f(x_1 + x_2) = 4(x_1 + x_2) + 2$$ Eq 3.6

Since Equations (3.5) and (3.6) are not identical, the additivity property does not hold.

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Linearity :

Informally,

a linear circuit is one in which the electronic components' values (such as resistance, capacitance, inductance, gain, etc.) do not change with the level of voltage or current in the circuit.

A circuit is linear if it consists only of:

- **Resistors (R)**
- **Inductors (L)**
- **Capacitors (C)**
- **Independent and dependent voltage/current sources (provided they follow linear relationships).**

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Linearity : **Example 3.1: Given the circuit shown in Figure 3.1. Use the concept of linearity (homogeneity or scaling) to find the current I_0.**

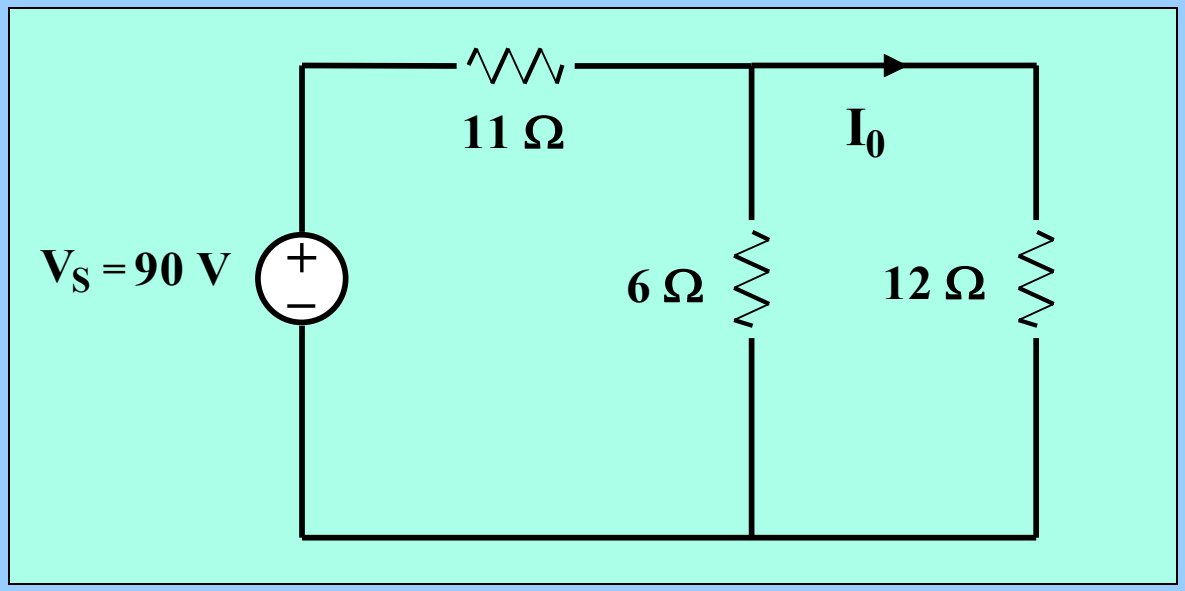


Figure 3.1: Circuit for Example 3.1.

Assume $I_0 = 1$ A. Work back to find that this gives $V_S = 45$ V. But since $V_S = 90$ V this means the true $I_0 = 2$ A.

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Linearity : **Example 3.2:** **In the circuit shown below it is known that $I_0 = 4$ A when $I_S = 6$ A. Find I_0 when $I_S = 18$ A.**

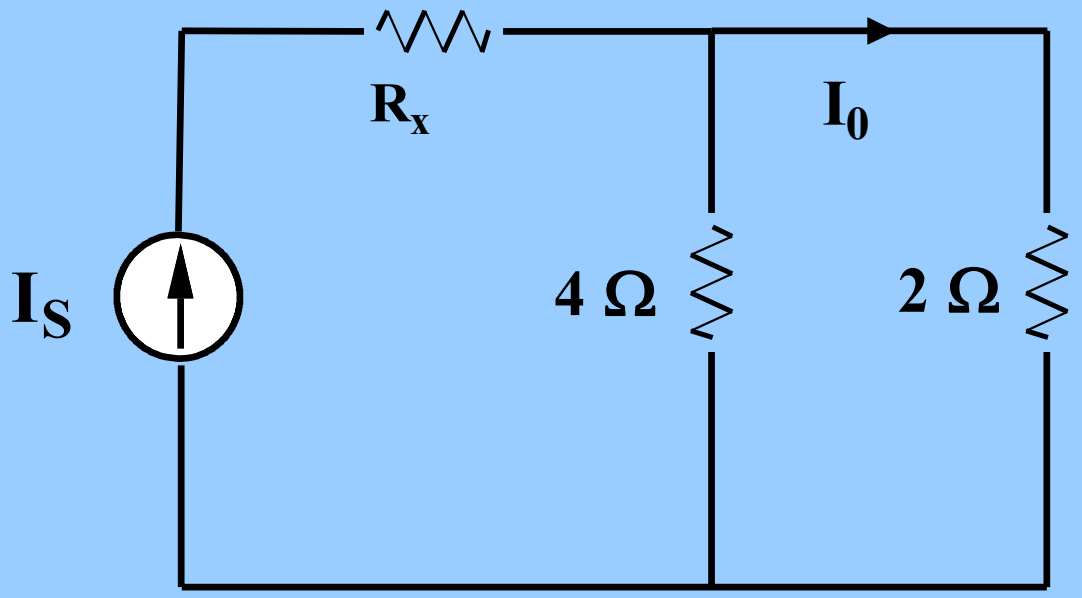


Figure 3.2: Circuit for Example 3.2

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Linearity : **Example 3.2:** **In the circuit shown below it is known that $I_0 = 4$ A when $I_S = 6$ A. Find I_0 when $I_S = 18$ A.**

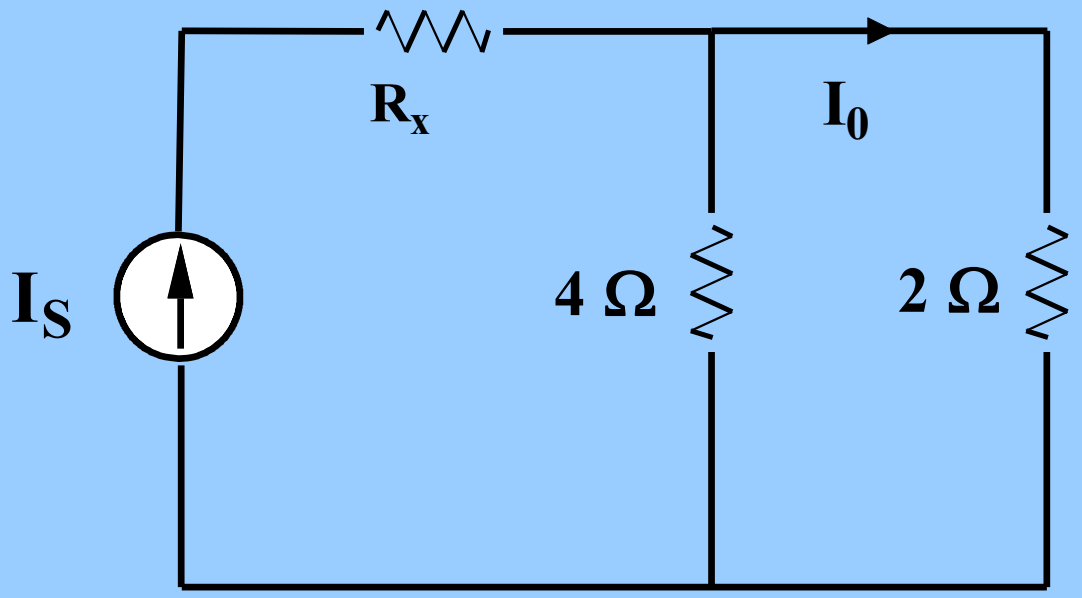


Figure 3.2: Circuit for Example 3.2

Since $I_{S\ NEW} = 3xI_{S\ OLD}$ we conclude $I_{0\ NEW} = 3xI_{0\ OLD}$.
Thus, $I_{0\ NEW} = 3x4 = 12$ A.

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Linearity : Question.

For Example 3.2, one might ask, "how do we know the circuit is linear?" That is a good question. To answer, we assume a circuit of the same form and determine if we get a linear equation between the output current and the input current. What must be shown for the circuit below?

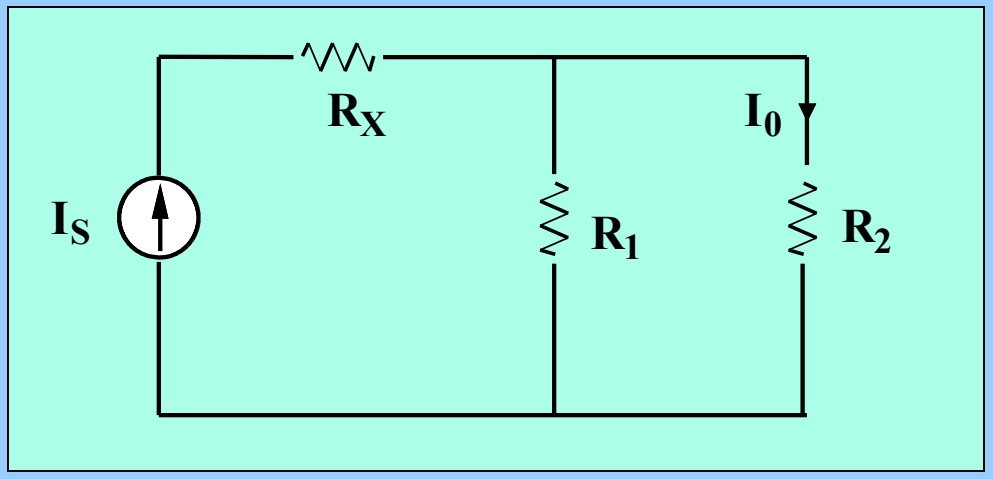


Figure 3.3: Circuit for investigating linearity.

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Linearity : **Question, continued.**

We use the current splitting rule (current division) to write the following equation.

$$I_0 = \frac{(I_S)(R_1)}{(R_1 + R_2)} = (K)(I_s) \qquad \text{Eq 3.7}$$

The equation is of the same form of y = mx, which we saw was linear. Therefore, if R_1 and R_2 are constants then the circuit is linear.

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Superposition :

One might read (hear) the following regarding superposition.

(1) A system is linear if superposition holds.

(2) Superposition holds if a system is linear.

This sounds a little like the saying of which comes first, “the chicken or the egg.”

Of the two statements, I believe one should remember that if a system is linear then superposition applies (holds).

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Superposition: Characterisation of Superposition.

Let inputs f_1 and f_2 be applied to a system y such that,

$$y = k_1 f_1 + k_2 f_2$$

Where k_1 and k_2 are constants of the systems.

Let f_1 act alone so that, $y = y_1 = k_1 f_1$
Let f_2 act alone so that, $y = y_2 = k_2 f_2$

The property of superposition states that if f_1 and f_2 Are applied together, the output y will be,

$$y = y_1 + y_2 = k_1 f_1 + k_2 f_2$$

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Superposition: Illustration using a circuit.

Consider the circuit below that contains two voltage sources.

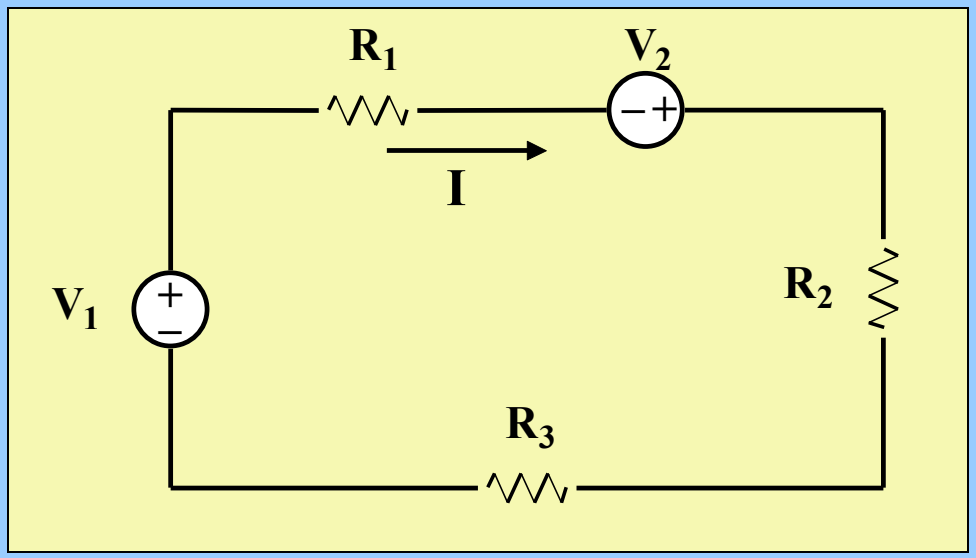
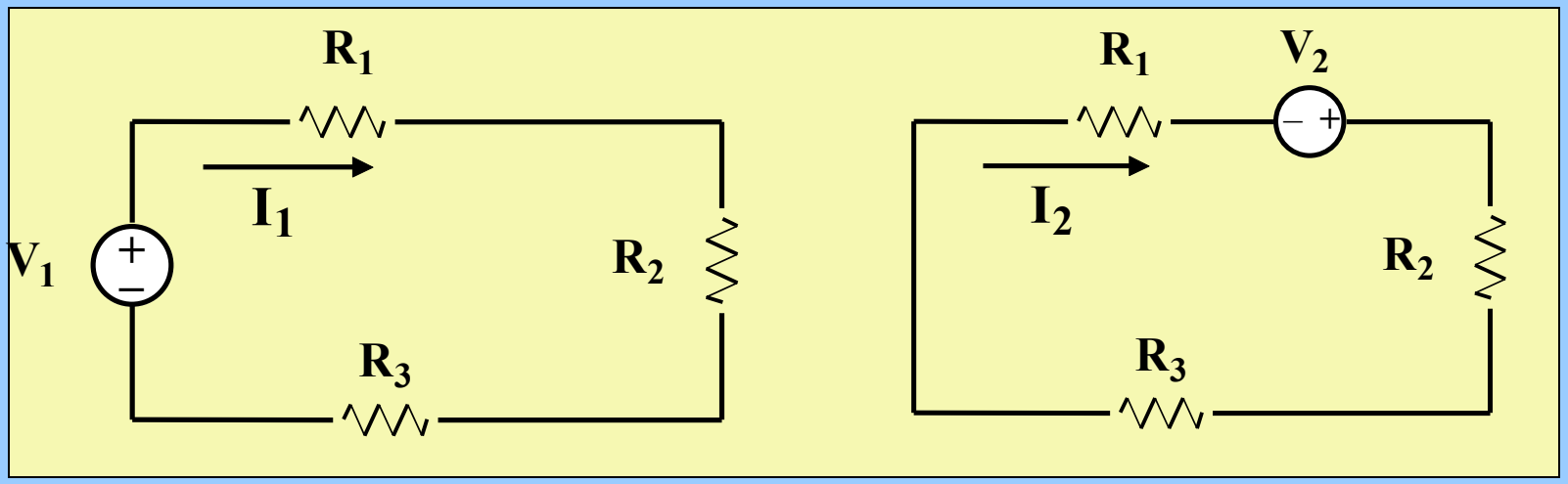


Figure 3.4: Circuit to illustrate superposition

We assume that V_1 and V_2 acting together produce current I.

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Superposition: Illustration using a circuit.



V_1 produces current I_1

V_2 produces current I_2

Superposition states that the current, I, produced by both sources acting together (Fig 3.4) is the same as the sum of the currents, $I_1 + I_2$, where I_1 is produced by V_1 and I_2 is produced by V_2.

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Superposition: **Example 3.3. Given the circuit below. Demonstration by solution that superposition holds.**

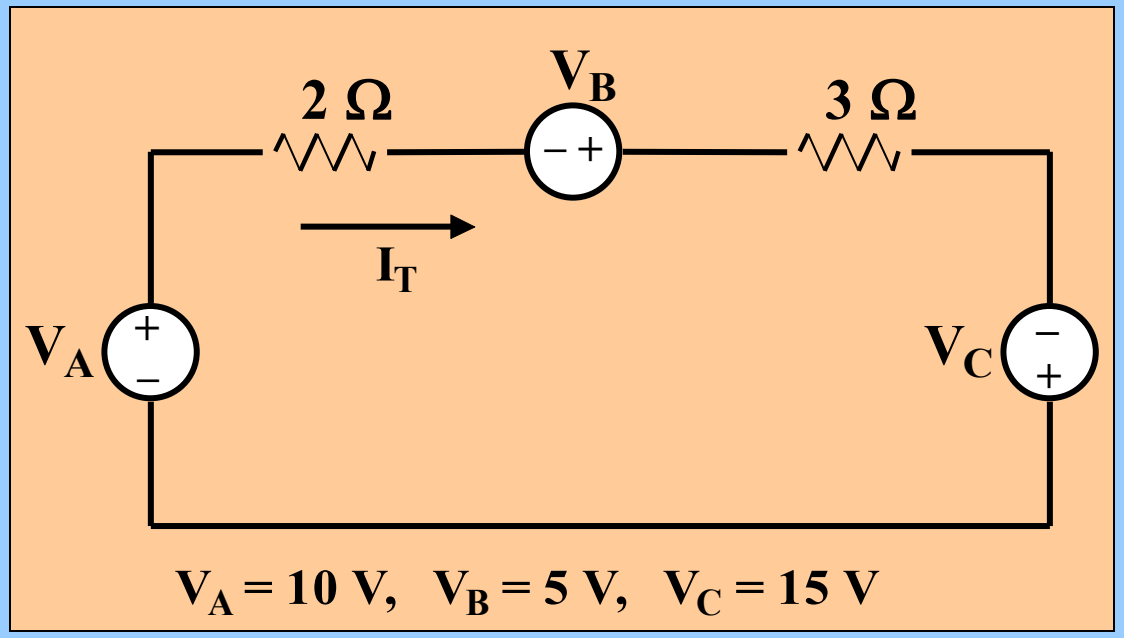
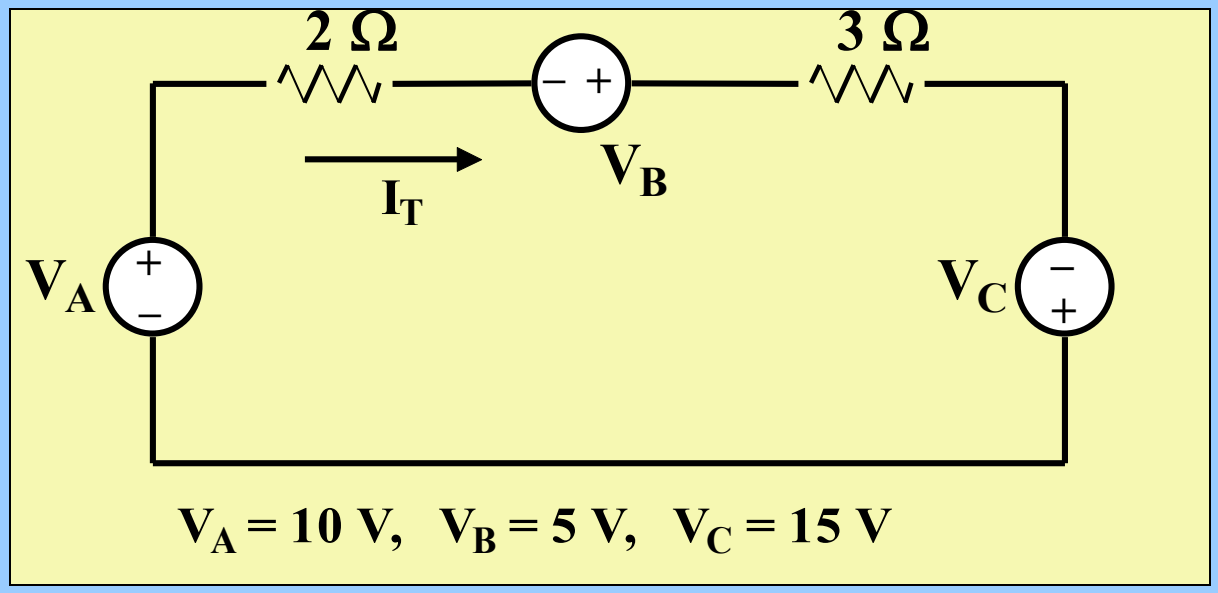


Figure 3.5: Circuit for Example 3.3

With all sources acting: $I_L = 6$ A

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Superposition: **Example 3.3. Given the circuit below. Demonstration by solution that superposition holds.**



With $V_A + V_B$ acting,,$V_C = 0$: $I_{A+B} = 3$ A

With V_C acting, $V_A + V_B = 0$: $I_C = 3$ A

We see that superposition holds.

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Superposition: **Example 3.4. Given the circuit below. Find the current I by using superposition.**

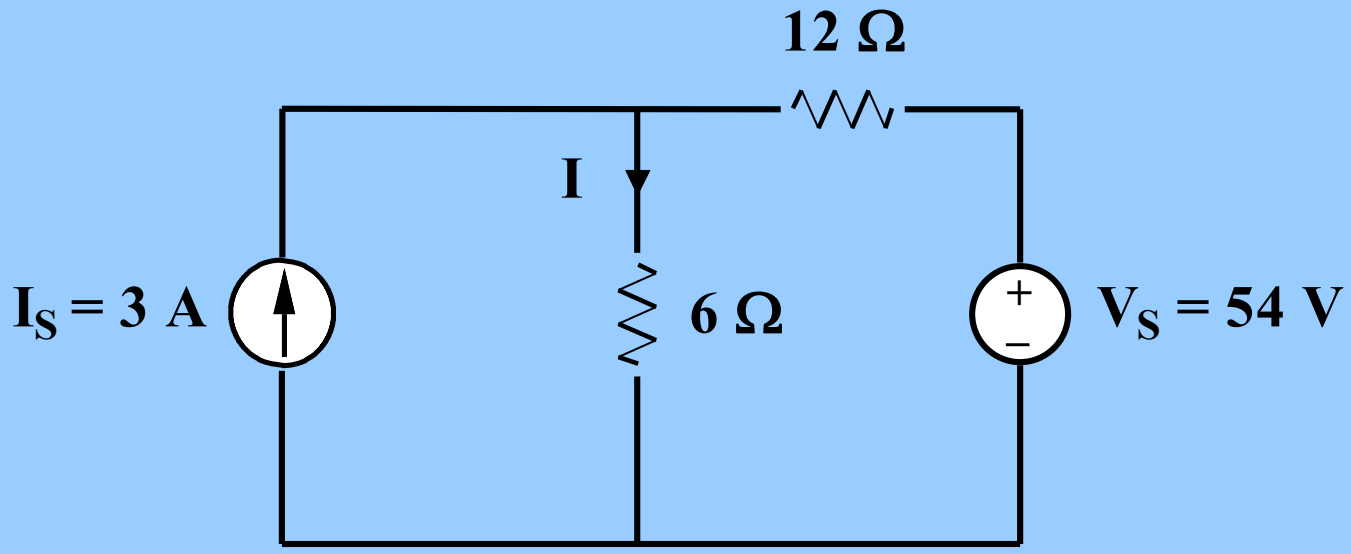


Figure 3.5: Circuit for Example 3.4.

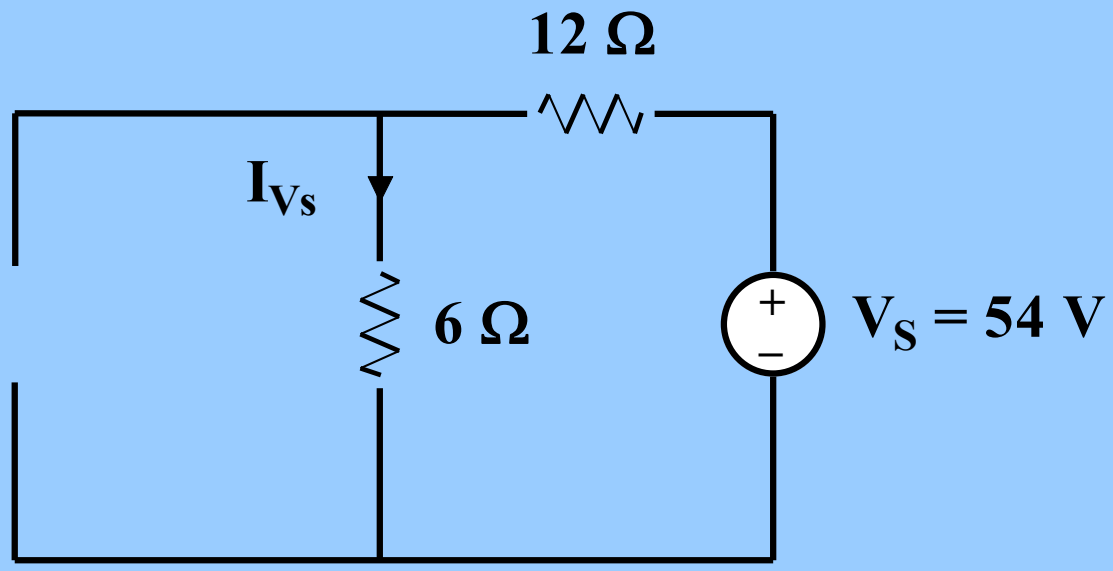
First, deactivate the source I_S and find I in the 6 Ω resistor.

Second, deactivate the source V_S and find I in the 6 Ω resistor.

Sum the two currents for the total current.

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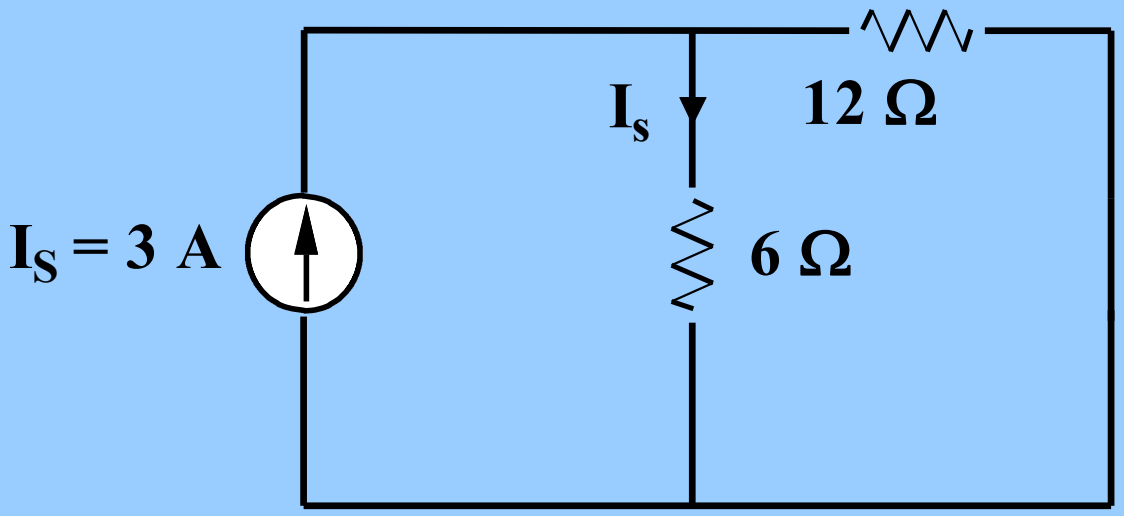
Superposition: **Example 3.4. Given the circuit below. Find the current I by using superposition.**



$I_{Vs} = 3$ A

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Superposition: **Example 3.4. Given the circuit below. Find the current I by using superposition.**



$$I_S = \frac{3x12}{(3+12)} = 2\,A$$

Total current I: $I = I_S + I_{vs} = 5$ A



End of Lesson

Linearity and Superposition